Q1.

Victoria and Albert play a zero-sum game. The game is represented by the following pay-off matrix for Victoria.

		Albert		
	Strategy	***X***	***Y***	***Z***
Victoria	***P***	3	–1	1
	Q	–2	0	1
	R	4	–1	–1

(a) Find the play-safe strategies for each player.

(3)

(b) State, with a reason, the strategy that Albert should never play.

(1)

(c) (i) Determine an optimal mixed strategy for Victoria.

(5)

(ii) Find the value of the game for Victoria.

(1)

(iii) State an assumption that must be made in order that your answer for part **(c)(ii)** is the maximum expected pay-off that Victoria can achieve.

(1)

(Total 11 marks)

Q2.

John and Danielle play a zero-sum game which does not have a stable solution.

The game is represented by the following pay-off matrix for John.

		Danielle		
	Strategy	***X***	***Y***	***Z***
John	***A***	2	1	–1
	B	–3	–2	2
	C	–3	–4	1

Find the optimal mixed strategy for John.

(Total 6 marks)

Q3.

Kate and Pippa play a zero-sum game. The game is represented by the following pay-off matrix for Kate.

Pippa

	Strategy	***D***	***E***	***F***
	A	−2	0	3
Kate	***B***	3	−2	−2
	C	4	1	−1

(a) Explain why Kate should not adopt strategy *B*.

(1)

(b) Find the optimal mixed strategy for Kate and find the value of the game.

(7)

(c) Find the optimal mixed strategy for Pippa.

(4)
(Total 12 marks)

Q4.

Romeo and Juliet play a zero-sum game. The game is represented by the following pay-off matrix for Romeo.

Juliet

	Strategy	**D**	**E**	**F**
	A	4	−4	0
Romeo	**B**	−2	−5	3
	C	2	1	−2

(a) Find the play-safe strategy for each player.

(3)

(b) Show that there is no stable solution.

(1)

(c) Explain why Juliet should never play strategy D.

(1)

(d) (i) Explain why the following is a suitable pay-off matrix **for Juliet**.

4	5	−1
0	−3	2

(2)

(ii) Hence find the optimal strategy for Juliet.

(7)

(iii) Find the value of the game for Juliet.

(1)

(Total 15 marks)

Q5.

Two people, Roz and Colum, play a zero-sum game. The game is represented by the following pay-off matrix for Roz.

		Colum		
	Strategy	**C_1**	**C_2**	**C_3**
	R_1	−2	−6	−1
Roz	**R_2**	−5	2	−6
	R_3	−3	3	−4

(a) Explain what is meant by the term 'zero-sum game'.

(2)

(b) Determine the play-safe strategy for Colum, giving a reason for your answer.

(2)

(c) (i) Show that the matrix can be reduced to a 2 by 3 matrix, giving the reason for deleting one of the rows.

(2)

(ii) Hence find the optimal mixed strategy for Roz.

(7)

(Total 13 marks)

Q6.

(a) Two people, Adam and Bill, play a zero-sum game. The game is represented by the following pay-off matrix for Adam.

		Bill		
	Strategy	**B_1**	**B_2**	**B_3**
	A_1	−6	−1	−5
Adam	**A_2**	5	2	−3
	A_3	−5	4	−4

A_4	2	1	−4

(i) Show that this game has a stable solution.

(3)

(ii) Find the play-safe strategy for each player.

(1)

(iii) State the value of the game for **Bill**.

(1)

(b) Roza plays a different zero-sum game against a computer. The game is represented by the following pay-off matrix for Roza.

Computer

	Strategy	**C_1**	**C_2**	**C_3**
	R_1	3	4	−3
Roza	**R_2**	−2	−1	5

(i) State which strategy the computer should never play, giving a reason for your answer.

(1)

(ii) Roza chooses strategy R_1 with probability p. Find expressions for the expected gains for Roza when the computer chooses each of its two remaining strategies.

(2)

(iii) Hence find the value of p for which Roza will maximise her expected gains.

(2)

(iv) Find the value of the game for Roza.

(1)

(Total 11 marks)

Q7.

Two people, Rhona and Colleen, play a zero-sum game. The game is represented by the following pay-off matrix for Rhona.

Colleen

	Strategy	**C_1**	**C_2**	**C_3**
	R_1	2	6	4
Rhona	**R_2**	3	−3	−1
	R_3	x	$x + 3$	3

It is given that $x < 2$.

(a) (i) Write down the three row minima.

(1)

(ii) Show that there is no stable solution.

(3)

(b) Explain why Rhona should never play strategy R_3.

(1)

(c) (i) Find the optimal mixed strategy for Rhona.

(7)

(ii) Find the value of the game.

(1)

(Total 13 marks)

Q8.

(a) Two people, Tom and Jerry, play a zero-sum game. The game is represented by the following pay-off matrix for Tom.

Jerry

	Strategy	**A**	**B**	**C**
	I	−4	5	−3
Tom	**II**	−3	−2	8
	III	−7	6	−2

Show that this game has a stable solution and state the play-safe strategy for each player.

(4)

(b) Rohan and Carla play a different zero-sum game for which there is no stable solution. The game is represented by the following pay-off matrix for Rohan.

Carla

	Strategy	C_1	C_2	C_3
Rohan	R_1	3	5	−1
	R_2	1	−2	4

(i) Find the optimal mixed strategy for Rohan and show that the value of the game is $\frac{3}{2}$.

(7)

(ii) Carla plays strategy C_1 with probability p, and strategy C_2 with probability q.

Find the values of p and q and hence find the optimal mixed strategy for Carla.

(4)

(Total 15 marks)

Q9.

(a) Two people, Ann and Bill, play a zero-sum game. The game is represented by the following pay-off matrix for Ann.

Bill

Ann

Strategy	**B_1**	**B_2**	**B_3**
A_1	–1	0	–2
A_2	4	–2	–3
A_3	–4	–5	–3

Show that this game has a stable solution and state the play-safe strategies for Ann and Bill.

(4)

(b) Russ and Carlos play a different zero-sum game, which does not have a stable solution. The game is represented by the following pay-off matrix for Russ.

Carlos

Russ

Strategy	**C_1**	**C_2**	**C_3**
R_1	–4	7	–3
R_2	2	–1	1

(i) Find the optimal mixed strategy for Russ.

(7)

(ii) Find the value of the game.

(1)

(Total 12 marks)

Q10.

Two people, Roger and Corrie, play a zero-sum game.

The game is represented by the following pay-off matrix for Roger.

Corrie

Roger

Strategy	**C_1**	**C_2**	**C_3**
R_1	7	3	–5
R_2	–2	–1	4

(a) (i) Find the optimal mixed strategy for Roger.

(7)

(ii) Show that the value of the game is $\frac{7}{13}$.

(1)

(b) Given that the value of the game is $\frac{7}{13}$, find the optimal mixed strategy for Corrie.

(5)

(Total 13 marks)

Q11.

(a) Two people, Raj and Cal, play a zero-sum game. The game is represented by the following pay-off matrix for Raj.

Cal

	Strategy	**X**	**Y**	**Z**
	I	–7	8	–5
Raj	**II**	6	2	–1
	III	–2	4	–3

Show that this game has a stable solution and state the play-safe strategy for each player.

(4)

(b) Ros and Carly play a different zero-sum game for which there is no stable solution. The game is represented by the following pay-off matrix for Ros, where x is a constant.

Carly

	Strategy	**C_1**	**C_2**
Ros	**R_1**	5	x
	R_2	–2	4

Ros chooses strategy R_1 with probability p.

(i) Find expressions for the expected gains for Ros when Carly chooses each of the strategies C_1 and C_2.

(2)

(ii) Given that the value of the game is $\frac{8}{3}$, find the value of p and the value of x.

(4)

(Total 10 marks)

Q12.

Two people, Rowena and Colin, play a zero-sum game.

The game is represented by the following pay-off matrix for Rowena.

Colin

	Strategy	C_1	C_2	C_3
	R_1	–4	5	4
Rowena	R_2	2	–3	–1
	R_3	–5	4	3

(a) Explain what is meant by the term 'zero-sum game'.

(1)

(b) Determine the play-safe strategy for Colin, giving a reason for your answer.

(2)

(c) Explain why Rowena should never play strategy R_3.

(1)

(d) Find the optimal mixed strategy for Rowena.

(7)

(Total 11 marks)

Q13.

Two people, Rob and Con, play a zero-sum game.

The game is represented by the following pay-off matrix for Rob.

Con

	Strategy	C_1	C_2	C_3
	R_1	–2	5	3
Rob	R_2	3	–3	–1
	R_3	–3	3	2

(a) Explain what is meant by the term 'zero-sum game'.

(1)

(b) Show that this game has no stable solution.

(3)

(c) Explain why Rob should never play strategy R_3.

(1)

(d) (i) Find the optimal mixed strategy for Rob.

(7)

(ii) Find the value of the game.

(1)
(Total 13 marks)

Q14.

Two people, Roseanne and Collette, play a zero-sum game. The game is represented by the following pay-off matrix for Roseanne.

		Collette		
	Strategy	**C_1**	**C_2**	**C_3**
Roseanne	**R_1**	–3	2	3
	R_2	2	–1	–4

(a) (i) Find the optimal mixed strategy for Roseanne.

(7)

(ii) Show that the value of the game is –0.5.

(1)

(b) (i) Collette plays strategy C_1 with probability p and strategy C_2 with probability q. Write down, in terms of p and q, the probability that she plays strategy C_3.

(1)

(ii) Hence, given that the value of the game is – 0.5, find the optimal mixed strategy for Collette.

(4)
(Total 13 marks)

Q15.

(a) Two people, Ros and Col, play a zero-sum game. The game is represented by the following pay-off matrix for Ros.

		Col		
	Strategy	**X**	**Y**	**Z**
Ros	**I**	–4	–3	0
	II	5	–2	2
	III	1	–1	3

(i) Show that this game has a stable solution.

(3)

(ii) Find the play-safe strategy for each player and state the value of the game.

(2)

(b) Ros and Col play a different zero-sum game for which there is no stable solution. The game is represented by the following pay-off matrix for Ros.

		Col		
	Strategy	**C_1**	**C_2**	**C_3**
Ros	**R_1**	3	2	1
	R_2	–2	–1	2

(i) Find the optimal mixed strategy for Ros.

(7)

(ii) Calculate the value of the game.

(1)

(Total 13 marks)

Q16.

Two people, Rose and Callum, play a zero-sum game. The game is represented by the following pay-off matrix for Rose.

		Callum		
		C_1	**C_2**	**C_3**
Rose	**R_1**	5	2	–1
	R_2	–3	–1	5
	R_3	4	1	–2

(a) (i) State the play-safe strategy for Rose and give a reason for your answer.

(2)

(ii) Show that there is no stable solution for this game.

(2)

(b) Explain why Rose should never play strategy **R_3**.

(1)

(c) Rose adopts a mixed strategy, choosing **R_1** with probability p and **R_2** with probability $1 - p$.

(i) Find expressions for the expected gain for Rose when Callum chooses each of his three possible strategies. Simplify your expressions.

(3)

(ii) Illustrate graphically these expected gains for $0 \leq p \leq 1$.

(2)

(iii) Hence determine the optimal mixed strategy for Rose.

(3)

(iv) Find the value of the game.

(1)

(Total 14 marks)

Q17.

Sam is playing a computer game in which he is trying to drive a car in different road conditions. He chooses a car and the computer decides the road conditions. The points scored by Sam are shown in the table.

Road Conditions

		C_1	C_2	C_3
Sam's Car	S_1	–2	2	4
	S_2	2	4	5
	S_3	5	1	2

Sam is trying to maximise his total points and the computer is trying to stop him.

(a) Explain why Sam should never choose S_1 and why the computer should not choose C_3.

(2)

(b) Find the play-safe strategies for the reduced 2 by 2 game for Sam and the computer, and hence show that this game does not have a stable solution.

(4)

(c) Sam uses random numbers to choose S_2 with probability p and S_3 with probability $1 - p$.

(i) Find expressions for the expected gain for Sam when the computer chooses each of its two remaining strategies.

(3)

(ii) Calculate the value of p for Sam to maximise his total points.

(2)

(iii) Hence find the expected points gain for Sam.

(1)

(Total 12 marks)

Q18.

Two people, Rowan and Colleen, play a zero-sum game. The game is represented by the following pay-off matrix for Rowan.

		Colleen		
	Strategy	C_1	C_2	C_3
	R_1	–3	–4	1
Rowan	R_2	1	5	–1
	R_3	–2	–3	4

(a) Explain the meaning of the term ‘zero-sum game’.

(1)

(b) Show that this game has no stable solution.

(3)

(c) Explain why Rowan should never play strategy R_1.

(1)

(d) (i) Find the optimal mixed strategy for Rowan.

(7)

(ii) Find the value of the game.

(1)

(Total 13 marks)

Q19.

Two people, Pat and Quigley, play a zero sum game. The game is represented by the following pay-off matrix for Pat.

		Quigley		
	Strategy	Q_1	Q_2	Q_3
Pat	P_1	4	3	5
	P_2	–1	5	–2
	P_3	1	2	3

(a) Show that there is no stable solution.

(3)

(b) Explain why Pat should never play strategy P_3.

(1)

(c) Find the optimal mixed strategy for Pat.

(7)

(Total 11 marks)

Q20.

Two people, A and B, play a zero-sum game. The game is represented by the following pay-off matrix for A.

		B		
	Strategy	**I**	**II**	**III**
	I	4	–1	2
A	**II**	2	3	1
	III	1	–2	1

(a) Explain why it will never be optimal for A to adopt strategy III.

(1)

(b) (i) Find the optimal mixed strategy for A.

(7)

(ii) Show that the value of the game is $\frac{7}{5}$.

(1)

(c) By considering optimal mixed strategies for B, show that player B should play strategy II with a probability of $\frac{1}{5}$.

(4)

(Total 13 marks)

Q21.

Arnie (A) and Ben (B) play a zero-sum game. The game is represented by the following pay-off matrix for Arnie (A).

		Ben (B)		
		I	II	III
Arnie (A)	I	1	2	2
	II	3	1	7
	III	2	3	6

(a) Explain why the matrix can be reduced to

3	1
2	3

(2)

(b) Find the optimal mixed strategy for each player and the value of the game.

(9)

(Total 11 marks)

Q22.

(a) Afzal and Bill play a zero-sum game. The game is represented by the following pay-off matrix for Afzal.

		Bill I	Bill II	Bill III	Bill IV
Afzal	I	3	–2	1	2
	II	3	5	2	6
	III	0	2	–3	3

(i) Show that this game has a stable solution.

(3)

(ii) List any saddle points.

(1)

(b) Colin and David playa zero-sum game. The game is represented by the following pay-off matrix for Colin.

		David I	David II
Colin	I	$x + 2$	$x - 1$
	II	3	5

Given that the optimal mixed strategy makes the value of the game $\frac{19}{5}$, find the value of x.

(8)

(Total 12 marks)

Q23.

Two people, Ann and Bev, play a zero sum game. The game is represented by the following pay-off matrix for Ann.

	Bev			
	Strategy	**I**	**II**	**III**
	I	1	3	6
Ann	**II**	6	1	1

III	7	4	2

(a) Show that there is no stable solution.

(3)

(b) Explain why it will never be optimal for Ann to adopt **Strategy II** and write down the reduced pay-off matrix for Ann.

(2)

(c) Hence, given that the value of the game is $3\frac{3}{5}$, find the optimal mixed strategy for **Bev**.

(*You are* ***not*** *required to find the optimal mixed strategy for Ann*.)

(5)

(Total 10 marks)

Q24.

Alan and Tracy play squash against each other every week. They each have three shots that they can play to try to win a point: a boast (B), a drop shot (D) and a lob (L). The pay-off matrix for Alan is given below.

Tracy

		B	D	L
Alan	B	12	5	2
	D	5	–2	–3
	L	8	10	13

(a) State which shot Alan should **not** play, giving a reason for your answer.

(2)

(b) By considering mixed strategies, find the ratio of the number of times Alan should play each of the other two shots.

(8)

(c) Find the value of the game.

(1)

(Total 11 marks)

Q25.

Two people, A and B, play a zero sum game. The game is represented by the following pay-off matrix for A.

B

Strategy	I	II	III

	I	5	1	3
A	II	2	5	4
	III	4	–1	2

(a) Show that there is no stable solution.

(2)

(b) Explain why it will never be optimal for A to adopt strategy III.

(1)

(c) By considering mixed strategies, and giving your answers as exact fractions,

(i) find the optimal mixed strategy for A,

(7)

(ii) find the value of the game.

(1)

(Total 11 marks)

Q26.

Alan (A) and Betty (B) play a zero-sum game. The game is represented by the following pay-off matrix for Alan (A).

		B I	B II	B III
A	I	5	2	7
A	II	4	1	5
A	III	3	5	6

(a) Explain why this matrix can be reduced to

5	2
3	5

(3)

(b) Find the optimal mixed strategy for each player and the value of the game.

(9)

(Total 12 marks)